

central median, supported by pylons separated by a distance of 30.5m (100 feet).

Thus no additional land would be required to set up this project. The series of pylons supporting the tube can be built over land that is already in use for public transportation, causing minimal interference with existing facilities. This also means significantly lower construction and operation costs than with any other mass transportation system in use.

Concerns

The idea of Hyperloop systems was first shared by Elon Musk in 2013. Even as Musk or SpaceX are working on the technology right now, many other companies such as Hyperloop Travel Technology (HTT) have picked up the concept and are trying to make the transportation system a reality.

The preliminary testing of the concept and design has been completed. If implemented successfully, the project, i.e., Elon Musk's Hyperloop, will become 'the fifth mode of transport' and the fastest in the world. This absolutely new type of high-speed transport is likely to connect the large cities of the world much more effectively than the high-speed railroads.

Designing and implementing an ultra-high-speed transportation system needs ultra-high computing power. For example, the HyperXite team is studying the fluid dynamics of the fuselage of this new vehicle.

Generally, a complete simulation requires over 900 CPU-hours. Earlier benchmarks performed by CycleCloud exhibited superb linearity with Microsoft Azure—up to and beyond 256 CPUs. This implies that executing on 256 cores completes the simulation approximately 256 times faster than on a single core.

There are some critics of Hyperloop who dwell more on the experience. They speculate that perhaps

it could be distasteful and terrifying to ride in a slender, sealed and windowless capsule, within a sealed steel tunnel. Besides, it would be exposed to considerably high acceleration forces, high noise levels due to air being compressed and ducted around the capsule at near-sonic speeds, and the vibration and tussle.

Even if the tube is smooth at the start of the journey, the ground could shift due to seismic activity. At such high speeds, even small deviations from a straight path could lead to significant buffeting. This is over and above issues and questions related to equipment malfunction, accidents and emergency evacuations.

Critics have also raised questions related to design technicalities in the tube system. Prof. John Hansman has highlighted the repercussions of a power blackout, especially when the pod is hundreds of kilometres away from a big city. Prof. Richard Muller has also raised fears about Hyperloop's novelty and defenselessness of its tubes. There is a high probability that Hyperloops could be targeted by terrorists.

However, all new inventions and breakthroughs had to face their share of fears, doubts and uncertainties. Yet, all great inventions changed the way we all live, work and play, for the better.

As per Elon Musk, Hyperloop is likely to prove very valuable on Mars as no tubes would be required there. Mars' atmosphere is only about one per cent the density of the Earth's. For the Hyperloop concept to work on the Earth, low-pressure tubes are needed to bring down air resistance. But on Mars, the considerably lower air resistance would allow a Hyperloop to be developed without a tube, since just a track would be sufficient. Perhaps several generations from now, those who settle in colonies on Mars, would take the Hyperloop to work! **EFY**

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